كلية تقنية المعلومات / قسم هندسة البرمجيات — الزمن : ساعتين

الامتحان النهائى لمادة البرمجة المرئية — الدرجة : 60

Q.1 Answer the following: **(12 Marks)**

1. **What is Python?**
2. **What are the uses of the Python language?**
3. **What are the features of the Python language?**

Q2. Predict the output for the following: **(20 Marks)**

1.
```
x = 5
y = 10
result = x ** y % y
print(result)  # Output?
```

2.
```
def multiply(a, b=2):
    return a * b
print(multiply(3))  # Output?
print(multiply(3, 4))  # Output?
```

3.
```
for i in range(3):
    for j in range(2):
        print(i, j)  # Output?
```

4.
```
arr = [1, 2, 3, 4]
print(arr[-1])  # Output?
print(arr[1:3])  # Output?
```

5.
```
student = {'name': 'Alice', 'age': 21, 'grade': 'A'}
print(student['name'])  # Output?
print(student.get('age'))  # Output?
print(student.get('gender', 'Not specified'))  # Output?
```

6.
```
a = [1, 2, 3, 4]
b = a
b.append(5)
print(a)  # Output?
print(b)  # Output?
```

7.
```
def foo(x):
    x[0] = 99
lst = [1, 2, 3]
foo(lst)
print(lst)  # Output?
```

8.
```
d = {"name": "John", "age": 30}
print("name" in d)  # Output?
print("John" in d)  # Output?
```

9.
```
x = 10
y = [x]
x = 20
print(y)  # Output?
print(x)  # Output?
```

10.
```
def bar():
    return "Python", "Exam"
a, b = bar()
print(a)  # Output?
print(b)  # Output?
```

Q3. Complete the Missing Code **(10 marks)**

1.
```
str1 = "hello"
str2 = "world"
print(result)  # Output: hello world
```

2.
```
def greet(name):
    print(f"Hello, {name}!")
  # Output: Hello, Alice!
```

3.
```
i = 0
while i <= 10:
  if i % 2 == 0:
      print(i)
```

4.
```
numbers = [1, 2, 3, 4]
print(numbers)  # Output: [1, 2, 3, 4, 5]
```

5.
```
person = {'name': 'Bob', 'weight': 70}
person['height'] = 170
# Output: {'name': 'Bob', 'weight': 70,
'height': 170}
```

Q.4 Write an algorithm and code to manage a contact list with the following functionalities: **(18 Marks)**

- **Add a new contact with a name and phone number.**
- **Delete a contact by name.**
- **Display all contacts in the contact list.**
- **Search for a contact by name and return the phone number.**